

thesis

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#Dataset:

```
#combine male&female:
colnames(df)[8] <- "Obesity_Prevalence"
colnames(df)[23] <- "Overweight_Prevalence"
combined<-aggregate(cbind(Obesity_Prevalence, Overweight_Prevalence) ~ Country.Region.World + Year + ISO,
combined["oao"]<-combined["Obesity_Prevalence"]+ combined["Overweight_Prevalence"]
countries<-c("EST", "FRA", "PER", "IND", "LVA", "COL", "IDN", "JPN", "LTU", "HUN", "SVN", "POL", "HRV",
result_oao<-subset(combined, ISO %in% countries)
```

#Direct medical costs Costs of 2016:

#Project oao of 2019:

```
# Fit a linear regression model
model <- lm(oao ~ Year + ISO, data = result_oao)
# Create a data frame with all combinations of ISO and Year for 2019
new_data <- expand.grid(Year = 2019, ISO = unique(result_oao$ISO))
# Make predictions for 2019
predicted_data <- data.frame(new_data, oao = predict(model, newdata = new_data))
predicted_data
```

##	Year	ISO	oao
## 1	2019	ARG	0.6588910
## 2	2019	AUS	0.6779810
## 3	2019	AUT	0.5907995
## 4	2019	BEL	0.6648614
## 5	2019	BGR	0.6623774
## 6	2019	BRA	0.5646388
## 7	2019	CAN	0.6675609
## 8	2019	CHE	0.5806910
## 9	2019	CHL	0.6598438
## 10	2019	CHN	0.3264188
## 11	2019	COL	0.6023550
## 12	2019	CRI	0.5822394
## 13	2019	CYP	0.6266420
## 14	2019	CZE	0.6971766
## 15	2019	DEU	0.6244718
## 16	2019	DNK	0.6090102
## 17	2019	ESP	0.6684005

```
## 18 2019 EST 0.6378554
## 19 2019 FIN 0.6244682
## 20 2019 FRA 0.6419099
## 21 2019 GBR 0.6636165
## 22 2019 GRC 0.6660743
## 23 2019 HRV 0.6344609
## 24 2019 HUN 0.6676106
## 25 2019 IDN 0.2828573
## 26 2019 IND 0.2445030
## 27 2019 IRL 0.6185180
## 28 2019 ISL 0.6471396
## 29 2019 ISR 0.7060745
## 30 2019 ITA 0.6358272
## 31 2019 JPN 0.3361139
## 32 2019 KOR 0.3521846
## 33 2019 LTU 0.6638658
## 34 2019 LUX 0.6291392
## 35 2019 LVA 0.6518636
## 36 2019 MEX 0.6631220
## 37 2019 MLT 0.7243530
## 38 2019 NLD 0.5991158
## 39 2019 NOR 0.6188596
## 40 2019 NZL 0.6866594
## 41 2019 PER 0.5950541
## 42 2019 POL 0.6386188
## 43 2019 PRT 0.5882607
## 44 2019 ROU 0.6157276
## 45 2019 RUS 0.6415242
## 46 2019 SAU 0.7145078
## 47 2019 SVK 0.6180345
## 48 2019 SVN 0.6095576
## 49 2019 SWE 0.6139108
## 50 2019 TUR 0.6605596
## 51 2019 USA 0.6925965
## 52 2019 ZAF 0.5336961
```

#linear regression of oao / mean oao in 2020-2050 globally

```
# Fit a linear regression model
model <- lm(oao ~ Year, data = result_oao)

# Predict the obesity prevalence for the years 2020 to 2050
years_to_predict <- data.frame(Year = 2020:2050)
predictions <- data.frame(
  Year = years_to_predict$Year,
  oao = predict(model, newdata = years_to_predict)
)

# Calculate the average obesity prevalence for the years 2020 to 2050
average_prevalence <- mean(predictions$oao)

average_prevalence
```

```
## [1] 0.6938953
```

#linear regression of oao on all the countries and use it to then predict medical costs in 2020-2050:

```
#prediction of every country:
projection <- data.frame()

for (country in unique(result_oao$Country.Region.World)){
  country_df<-result_oao[result_oao$Country.Region.World == country, ]

  model <- lm(oao ~ Year, data = country_df)
  years_to_predict <- data.frame(Year = 2020:2050)
  predictions <- data.frame(
    Year = years_to_predict$Year,
    oao = predict(model, newdata = years_to_predict)
  )

  # Calculate the average obesity prevalence for the years 2020 to 2050
  average_prevalence <- mean(predictions$oao)
  average_prevalence

  # Store the results in the 'results' data frame
  projection <- rbind(projection, data.frame(Country = country, Average_Prevalence = average_prevalence))
}

# Print the results
print(projection)
```

```
##           Country Average_Prevalence
## 1      Argentina      0.7598624
## 2      Australia      0.7806895
## 3        Austria      0.6600606
## 4        Belgium      0.6991962
## 5        Bulgaria      0.7312345
## 6         Brazil      0.7297490
## 7         Canada      0.7850908
## 8      Switzerland      0.7001598
## 9          Chile      0.7621448
## 10         China      0.4233123
## 11       Colombia      0.7448922
## 12      Costa Rica      0.7957963
## 13         Cyprus      0.7546971
## 14   Czech Republic      0.7105984
## 15         Germany      0.6814683
## 16         Denmark      0.6711368
## 17          Spain      0.7406366
## 18         Estonia      0.6323507
## 19         Finland      0.7248863
## 20          France      0.7098977
## 21   United Kingdom      0.7734308
## 22          Greece      0.7488377
## 23         Croatia      0.7155573
```

## 24	Hungary	0.7101168
## 25	Indonesia	0.3741492
## 26	India	0.2584024
## 27	Ireland	0.7588818
## 28	Iceland	0.7089509
## 29	Israel	0.7559549
## 30	Italy	0.6954442
## 31	Japan	0.3277005
## 32	South Korea	0.4122670
## 33	Lithuania	0.7029460
## 34	Luxembourg	0.7122445
## 35	Latvia	0.6635311
## 36	Mexico	0.8078585
## 37	Malta	0.7681735
## 38	Netherlands	0.7254092
## 39	Norway	0.7143758
## 40	New Zealand	0.7965324
## 41	Peru	0.7101019
## 42	Poland	0.6854314
## 43	Portugal	0.7298414
## 44	Romania	0.6987851
## 45	Russian Federation	0.6529500
## 46	Saudi Arabia	0.8843285
## 47	Slovakia	0.6769389
## 48	Slovenia	0.6741729
## 49	Sweden	0.6740609
## 50	Turkey	0.8606604
## 51	United States of America	0.8398900
## 52	South Africa	0.6667720

```
#expand the model/regress it on direct medial costs so we can always plug in a number for obesity preve
projection<-projection[order(projection$Average_Prevalence),]
print(projection)
```

##	Country	Average_Prevalence
## 26	India	0.2584024
## 31	Japan	0.3277005
## 25	Indonesia	0.3741492
## 32	South Korea	0.4122670
## 10	China	0.4233123
## 18	Estonia	0.6323507
## 45	Russian Federation	0.6529500
## 3	Austria	0.6600606
## 35	Latvia	0.6635311
## 52	South Africa	0.6667720
## 16	Denmark	0.6711368
## 49	Sweden	0.6740609
## 48	Slovenia	0.6741729
## 47	Slovakia	0.6769389
## 15	Germany	0.6814683
## 42	Poland	0.6854314
## 30	Italy	0.6954442
## 44	Romania	0.6987851
## 4	Belgium	0.6991962

##	8	Switzerland	0.7001598
##	33	Lithuania	0.7029460
##	28	Iceland	0.7089509
##	20	France	0.7098977
##	41	Peru	0.7101019
##	24	Hungary	0.7101168
##	14	Czech Republic	0.7105984
##	34	Luxembourg	0.7122445
##	39	Norway	0.7143758
##	23	Croatia	0.7155573
##	19	Finland	0.7248863
##	38	Netherlands	0.7254092
##	6	Brazil	0.7297490
##	43	Portugal	0.7298414
##	5	Bulgaria	0.7312345
##	17	Spain	0.7406366
##	11	Colombia	0.7448922
##	22	Greece	0.7488377
##	13	Cyprus	0.7546971
##	29	Israel	0.7559549
##	27	Ireland	0.7588818
##	1	Argentina	0.7598624
##	9	Chile	0.7621448
##	37	Malta	0.7681735
##	21	United Kingdom	0.7734308
##	2	Australia	0.7806895
##	7	Canada	0.7850908
##	12	Costa Rica	0.7957963
##	40	New Zealand	0.7965324
##	36	Mexico	0.8078585
##	51	United States of America	0.8398900
##	50	Turkey	0.8606604
##	46	Saudi Arabia	0.8843285

```
projection["oafs"]=c(5.5834, 6.1217, 7.5428, 7.7690, 6.0866, 4.6819, 9.1332, 9.0287, 5.5180, 6.4475, 8.1111)
projection
```

##	Country	Average_Prevalence	oafs
## 26	India	0.2584024	5.5834
## 31	Japan	0.3277005	6.1217
## 25	Indonesia	0.3741492	7.5428
## 32	South Korea	0.4122670	7.7690
## 10	China	0.4233123	6.0866
## 18	Estonia	0.6323507	4.6819
## 45	Russian Federation	0.6529500	9.1332
## 3	Austria	0.6600606	9.0287
## 35	Latvia	0.6635311	5.5180
## 52	South Africa	0.6667720	6.4475
## 16	Denmark	0.6711368	8.2022
## 49	Sweden	0.6740609	9.0900
## 48	Slovenia	0.6741729	6.9060
## 47	Slovakia	0.6769389	6.6760
## 15	Germany	0.6814683	11.3370
## 42	Poland	0.6854314	7.6690

```
## 30          Italy          0.6954442  4.8780
## 44          Romania       0.6987851  9.3930
## 4           Belgium       0.6991962 10.0650
## 8           Switzerland   0.7001598  6.0080
## 33          Lithuania     0.7029460  8.5360
## 28          Iceland       0.7089509 10.7330
## 20          France        0.7098977  6.4490
## 41          Peru          0.7101019  6.9600
## 24          Hungary       0.7101168  6.0220
## 14          Czech Republic 0.7105984  8.3760
## 34          Luxembourg     0.7122445  8.6910
## 39          Norway        0.7143758  9.3540
## 23          Croatia       0.7155573  9.7430
## 19          Finland       0.7248863 10.1240
## 38          Netherlands   0.7254092  9.7140
## 6           Brazil        0.7297490  6.6480
## 43          Portugal       0.7298414  7.0630
## 5           Bulgaria       0.7312345  8.4700
## 17          Spain         0.7406366  6.2460
## 11          Colombia      0.7448922  9.1400
## 22          Greece        0.7488377  6.1950
## 13          Cyprus        0.7546971  8.0720
## 29          Israel        0.7559549  9.0330
## 27          Ireland       0.7588818  6.9560
## 1           Argentina     0.7598624  7.5340
## 9           Chile         0.7621448 10.3450
## 37          Malta         0.7681735  8.6130
## 21          United Kingdom 0.7734308  8.3560
## 2           Australia     0.7806895  9.9100
## 7           Canada        0.7850908  8.5860
## 12          Costa Rica    0.7957963  8.9340
## 40          New Zealand    0.7965324 10.6280
## 36          Mexico        0.8078585  7.9200
## 51 United States of America 0.8398900 11.5760
## 50          Turkey        0.8606604 13.5310
## 46          Saudi Arabia   0.8843285 12.6520
```

```
final<-lm(projection$Average_Prevalence~projection$oaafs)
final
```

```
##
## Call:
## lm(formula = projection$Average_Prevalence ~ projection$oaafs)
##
## Coefficients:
##      (Intercept)  projection$oaafs
##           0.4438             0.0303
```

```
final$coefficients%*%c(1,13)
```

```
##           [,1]
## [1,] 0.8376633
```

```
summary(final)
```

```
##
## Call:
## lm(formula = projection$Average_Prevalence ~ projection$oaafs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.35456 -0.02398  0.02918  0.07105  0.12411
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.443799   0.067025   6.621 2.35e-08 ***
## projection$oaafs 0.030297   0.007909   3.831 0.000358 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1093 on 50 degrees of freedom
## Multiple R-squared:  0.2269, Adjusted R-squared:  0.2114
## F-statistic: 14.67 on 1 and 50 DF,  p-value: 0.0003576
```

```
#should be correct:
```

```
final<-lm(projection$oaafs~projection$Average_Prevalence)
final
```

```
##
## Call:
## lm(formula = projection$oaafs ~ projection$Average_Prevalence)
##
## Coefficients:
##              (Intercept)  projection$Average_Prevalence
##                   3.058                        7.489
```

```
final$coefficients%*%c(1,0.5)
```

```
##           [,1]
## [1,] 6.802691
```

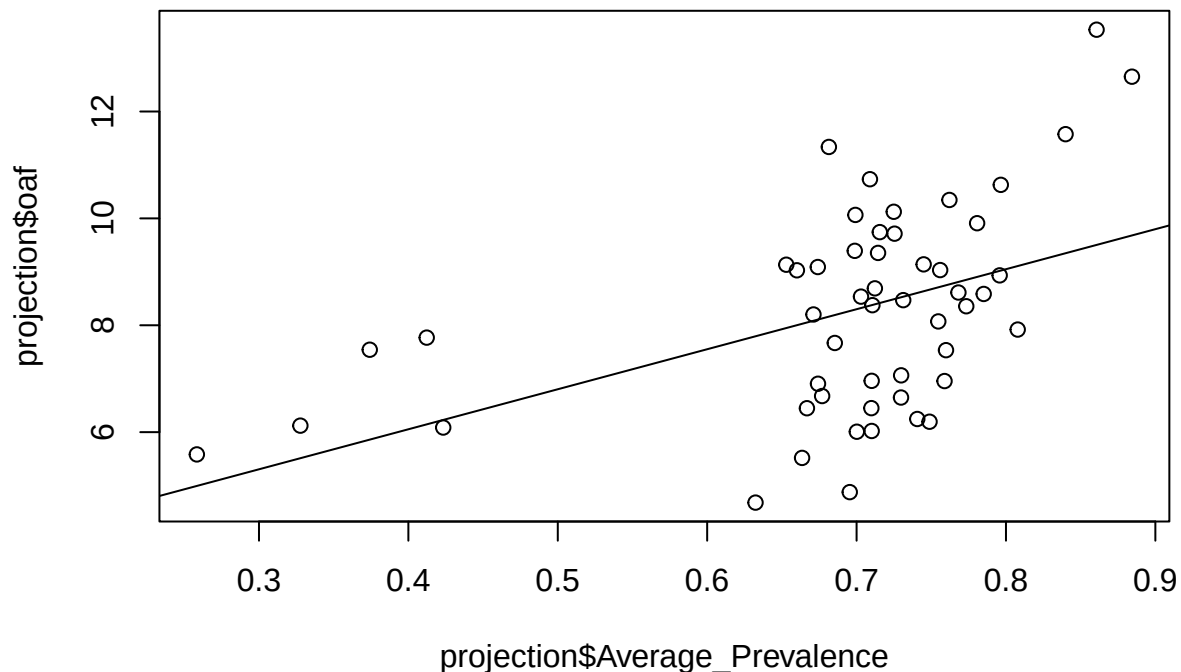
```
summary(final)
```

```
##
## Call:
## lm(formula = projection$oaafs ~ projection$Average_Prevalence)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.388 -1.425  0.057  1.195  4.027
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.058     1.377   2.220 0.030951 *
```

```
## projection$Average_Prevalence    7.489      1.955    3.831 0.000358 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.719 on 50 degrees of freedom
## Multiple R-squared:  0.2269, Adjusted R-squared:  0.2114
## F-statistic: 14.67 on 1 and 50 DF,  p-value: 0.0003576
```

#why is my coefficient different to the one in the paper?

```
plot(projection$ofaf~projection$Average_Prevalence)
abline(final$coefficients)
```



#direct non-medical costs: ##travel cost:

```
gas<-read.csv("e2373858-926a-4755-8350-78bb920f6ee9_Data.csv")
gas <- gas[, c(1,2,3,4,50,51,52,53,54,55,56,57,58, 59, 60, 61)]
gas$gas2016<-gas$X2016..YR2016.
#replace missing values by mean:
gas$gas2016[is.na(gas$gas2016)] <- NA
m <- mean(as.numeric(gas$gas2016), na.rm = TRUE)
gas$gas2016[is.na(gas$gas2016)] <- m
#reduce it to countries used above:
countries<-c("EST", "FRA", "PER", "IND", "LVA", "COL", "IDN", "JPN", "LTU", "HUN", "SVN", "POL", "HRV",

result_gas<-subset(gas, Country.Code %in% countries)
head(result_gas)
```

##	Country.Name	Country.Code	Series.Name		
## 2	Australia	AUS	Pump price for gasoline (US\$ per liter)		
## 3	Austria	AUT	Pump price for gasoline (US\$ per liter)		
## 4	United States	USA	Pump price for gasoline (US\$ per liter)		
## 14	Argentina	ARG	Pump price for gasoline (US\$ per liter)		
## 23	Belgium	BEL	Pump price for gasoline (US\$ per liter)		
## 31	Brazil	BRA	Pump price for gasoline (US\$ per liter)		
##	Series.Code	X2005..YR2005.	X2006..YR2006.	X2007..YR2007.	X2008..YR2008.
## 2	EP.PMP.SGAS.CD	..	0.93	..	0.74
## 3	EP.PMP.SGAS.CD	..	1.32	..	1.37
## 4	EP.PMP.SGAS.CD	..	0.63	..	0.56
## 14	EP.PMP.SGAS.CD	..	0.62	..	0.78
## 23	EP.PMP.SGAS.CD	..	1.63	..	1.5
## 31	EP.PMP.SGAS.CD	..	1.26	..	1.26
##	X2009..YR2009.	X2010..YR2010.	X2011..YR2011.	X2012..YR2012.	X2013..YR2013.
## 2	..	1.27	..	1.39	..
## 3	..	1.63	..	1.81	..
## 4	..	0.76	..	0.97	..
## 14	..	0.96	..	1.46	..
## 23	..	1.87	..	2.09	..
## 31	..	1.58	..	1.39	..
##	X2014..YR2014.	X2015..YR2015.	X2016..YR2016.	gas2016	
## 2	1.23	..	0.93	0.93	
## 3	1.6	..	1.2	1.2	
## 4	0.76	..	0.71	0.71	
## 14	1.52	..	1.1	1.1	
## 23	1.9	..	1.43	1.43	
## 31	1.27	..	1.02	1.02	

```

population<-read.csv("population_WHO.csv")
population <- population[, c(1,2,3,4,50,51,52,53,54,55,56,57,58, 59, 60, 61)]

#reduce it to countries used above:
result_population<-subset(population, Country.Code %in% countries)
head(result_population)

```

##	Country.Name	Country.Code	Series.Name	Series.Code	X2005..YR2005.
## 8	Argentina	ARG	Population, total	SP.POP.TOTL	39070501
## 11	Australia	AUS	Population, total	SP.POP.TOTL	20176844
## 12	Austria	AUT	Population, total	SP.POP.TOTL	8227829
## 19	Belgium	BEL	Population, total	SP.POP.TOTL	10478617
## 27	Brazil	BRA	Population, total	SP.POP.TOTL	186797334
## 30	Bulgaria	BGR	Population, total	SP.POP.TOTL	7658972
##	X2006..YR2006.	X2007..YR2007.	X2008..YR2008.	X2009..YR2009.	X2010..YR2010.
## 8	39476851	39876111	40273769	40684338	40788453
## 11	20450966	20827622	21249199	21691653	22031750
## 12	8268641	8295487	8321496	8343323	8363404
## 19	10547958	10625700	10709973	10796493	10895586
## 27	188820682	190779453	192672317	194517549	196353492
## 30	7601022	7545338	7492561	7444443	7395599
##	X2011..YR2011.	X2012..YR2012.	X2013..YR2013.	X2014..YR2014.	X2015..YR2015.
## 8	41261490	41733271	42202935	42669500	43131966
## 11	22340024	22733465	23128129	23475686	23815995
## 12	8391643	8429991	8479823	8546356	8642699

```
## 19      11038264      11106932      11159407      11209057      11274196
## 27      198185302      199977707      201721767      203459650      205188205
## 30      7348328      7305888      7265115      7223938      7177991
##      X2016..YR2016.
## 8      43590368
## 11      24190907
## 12      8736668
## 19      11331422
## 27      206859578
## 30      7127822
```

```
#average outpatient visits per capita
outpatient<-5.42
#average inpatient visits per capita
inpatient<-0.10
```

calculate for 2016:

```
#we first have to order all datasets so the used values match the corresponding country:
result_gas<-result_gas[order(result_gas$Country.Code), ]
result_population<-result_population[order(result_population$Country.Code),]
result_oao_2016<-result_oao[result_oao$Year == "2016", ]
result_oao_2016<-result_oao_2016[order(result_population$Country.Code),]
result_oao_2016$population_with_obesity<-as.numeric(result_population$X2016..YR2016.)*as.numeric(result_oao_2016$population_with_obesity)

#gas price per l * km * l/km:
result_gas$atc<-as.numeric(result_gas$gas2016)*5*0.06

outpatient_travel_cost<-result_gas$atc*outpatient*result_oao_2016$population_with_obesity
inpatient_travel_cost<-result_gas$atc*inpatient*result_oao_2016$population_with_obesity
outpatient_travel_cost
```

```
## [1] 50309839.7 24261569.4 9527511.8 16136820.3 8160516.4 199886913.7
## [7] 31355448.0 11032820.2 19617905.1 722242468.1 31960913.5 4989092.4
## [13] 1452287.5 12876645.8 108781505.4 8241818.7 60289371.4 1403886.2
## [19] 7714024.9 92469655.7 102119747.2 17257927.0 5254194.1 11955993.0
## [25] 78135110.4 432174082.3 6611648.9 558927.7 14425575.7 95534191.4
## [31] 61469902.0 31640509.9 3318381.9 679490.1 2202884.8 95944694.1
## [37] 684954.4 27665537.3 9070119.2 7240705.9 29578297.8 39779233.1
## [43] 15332376.7 22107225.6 81471100.4 9380528.6 6759625.4 2557851.3
## [49] 13288930.2 128469603.7 259876643.7 46068654.1
```

```
inpatient_travel_cost
```

```
## [1] 928225.82 447630.43 175784.35 297727.31 150563.03 3687950.44
## [7] 578513.80 203557.57 361953.97 13325506.79 589684.75 92049.68
## [13] 26794.97 237576.49 2007038.85 152063.08 1112350.03 25901.96
## [19] 142325.18 1706082.21 1884128.18 318411.94 96940.85 220590.28
## [25] 1441607.20 7973691.56 121986.14 10312.32 266154.53 1762623.46
## [31] 1134131.03 583773.25 61224.76 12536.72 40643.63 1770197.31
## [37] 12637.54 510434.27 167345.37 133592.36 545725.05 733934.19
## [43] 282885.18 407882.39 1503156.83 173072.48 124716.34 47192.83
## [49] 245183.21 2370287.89 4794772.02 849975.17
```

```
travel_cost<-outpatient_travel_cost+inpatient_travel_cost
travel_cost
```

```
## [1] 51238065.5 24709199.8 9703296.1 16434547.6 8311079.4 203574864.1
## [7] 31933961.8 11236377.8 19979859.1 735567974.8 32550598.3 5081142.1
## [13] 1479082.4 13114222.3 110788544.3 8393881.8 61401721.5 1429788.2
## [19] 7856350.1 94175737.9 104003875.3 17576338.9 5351134.9 12176583.3
## [25] 79576717.6 440147773.9 6733635.0 569240.0 14691730.2 97296814.9
## [31] 62604033.0 32224283.2 3379606.6 692026.8 2243528.4 97714891.4
## [37] 697592.0 28175971.6 9237464.6 7374298.3 30124022.9 40513167.3
## [43] 15615261.8 22515108.0 82974257.2 9553601.1 6884341.7 2605044.2
## [49] 13534113.4 130839891.6 264671415.7 46918629.3
```

##Informal caregiver travel cost

```
caregiver_travel_cost<-result_gas$atc*inpatient*result_oao_2016$population_with_obesity
total_travel_cost<-caregiver_travel_cost+travel_cost
```

#indirect costs: